

Please check that this question paper contains 9 questions and 2 printed pages within first ten minutes.

EVENING

[Total No. of Questions: 09]

[Total No. of Pages: 02]

29 MAY 2024

Uni. Roll No.

Program/ Course: B.Tech (Batch 2018 onward)

Semester: 3rd

Name of Subject: Mathematics-III

Subject Code: BSCS-101

Paper ID: 16013

Scientific Calculator is Allowed

Time Allowed: 3 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory.
- 2) Part -C has Two questions Q8 and Q9. Both are compulsory, but with internal choice.
- 3) Any missing data may be assumed appropriately.

Part -A

[Marks: 02 each]

Q1.

- a) Separate $f(z) = \frac{1}{z}$ into real and imaginary parts.
- b) Solve the system of equations: $3x - 2y = 1$, $x + 2y = 7$
- c) Evaluate $\int_C \frac{dz}{(z+2)^2}$ where C is the circle $|z+i|=2$.
- d) Write algorithm for straight line fit of a curve.
- e) Check the correctness of the statement, "Mean of a binomial distribution is 9 and S.D is $3/2$."
- f) Write Cauchy Riemann equations in cartesian form.

Part -B

[Marks: 04 each]

Q2. Fit the curve $y = a + bx$ to the following data :

x	1	2	3	4	5
y	14	27	40	55	68

Q3. Expand $f(z) = \frac{1}{(z-1)(z-2)}$ in the region $1 < |z| < 2$.

Q4. Solve the following equations by Gauss Elimination method :

$$2x - 2y + z = 1, \quad x + 2y + 2z = 2, \quad 2x + y - 2z = 7$$

Q5. A random sample of 9 items had the following values:

45, 47, 50, 51, 48, 47, 49, 53, 51. Does the mean of these values differ significantly from the assumed mean 47.5 at 5% level of significance? (Tabulated value = 2.31)

Q6. If on an average 1 vessel in every 10 is wrecked, find the probability that out of 5 vessels expected to arrive, at least 4 will arrive safely.

Q7. If $f(z) = u + iv$ is an analytic function and $u = e^x(\cos y - \sin y)$, find $f(z)$.

Part -C

[Marks: 12 each]

Q8. Evaluate $\int_0^{2\pi} \frac{d\theta}{(5-3\cos\theta)}$ using contour integration.

OR

Evaluate $\int_C \frac{dz}{z^2(z+1)(z-1)}$, where C is $|z| = 3$, using Cauchy Residue theorem.

Q9. Fit a Poisson distribution to the following data and test the goodness of fit :

x	0	1	2	3	4
f	122	60	15	2	1

(Tabulated value = 7.815)

OR

A random variable X has the following probability distribution :

X	0	1	2	3	4	5	6	7	8
P(X)	a	3a	5a	7a	9a	11a	13a	15a	17a

- Find a .
- Find $P(2 \leq X < 5)$.
- Mean of X .
